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StructureLDM: A Latent Diffusion Model for Sketch-based Structural Analysis

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Abstract

In the early stage of sketch-based product design for digital fabrication, rapid structural analysis serves as an efficient way to analyze the weakness and optimize the structure of the designed product and is widely adopted in agile prototyping. However, performing sketch-based structural analysis during the design phase is challenging due to the abstract nature of sketches, improper representation of external forces, and insufficient analysis models. Existing sketch-based structural analysis methods typically rely on simple mapping from input sketches and force maps to output stress maps in the generation procedure, suffering from limited analysis quality due to poor spatial control of external forces. To address these challenges, we reformulate the sketch-based structural analysis problem as a conditional generation task and propose StructureLDM, a novel framework for sketch-based structural analysis that leverages the latent-diffusion model as the generative engine for the analytical stress map synthesis. Our StructureLDM takes as input a sketch and the coordinates of the external force applied at the target position on the sketch. With a specifically-designed force-point embedding module for precise spatial force control and a Fourier-domain loss for the high-frequency contexts enhancement, it automatically predicts an accurate stress map. Comprehensive qualitative and quantitative experiments show the effectiveness of our proposed method, significantly surpassing previous approaches in terms of fine-grained spatial force controllability and high-precision stress map generation.

CCS Concepts

• ***Computing methodologies*** → *Image processing; Graphics systems and interfaces;*

1. Introduction

The product design and digital fabrication pipeline typically begins with the conceptual sketching, enabling designers to explore and express ideas intuitively and creatively. Traditionally, such procedure involves digital modeling and prototype fabrication using 3D printing, additive manufacturing, and laser cutting, and concludes with structural analysis to assess whether the designed structure is able to withstand the external forces [BGW23, SLZ*13, YXLF24]. The traditional process is resource-intensive, requiring considerable time, labor, and materials. To facilitate product design and digital fabrication, structural analysis techniques [SVB*12, ZPZ13, UMK17, PRZ17] have been proposed to simulate the physical environment and enable direct analysis or optimization of digital prototype structures virtually at the digitization stage. These techniques can be roughly classified into the following aspects: weakness analysis [TR21, MR09, SVB*12, ZPZ13], topology optimization [PRZ17, FCK26], and material distribution optimization [UMK17, ZCT16, LSZ*14] on parametric digital models. While such structural analysis tools significantly enrich product design and fabrication during the digital modeling stage on detailed 3D models, they are less accessible to designers at the early sketching stage, where the effect of external force on the designed object is unknown to them.

In this work, we formulate the process as sketch-based structural analysis, which takes as input a sketched object together with specified external forces and predicts the corresponding stress distribution over the designed object (stress map), as illustrated in Figure 1. Addressing this problem enables users to examine and analyze the structural design (e.g., stress patterns, weak regions) of a object directly from its sketch, without requiring a full 3D CAD model or a traditional computer-aided engineering (CAE) analysis pipeline that is often inaccessible and demanding for ordinary users.

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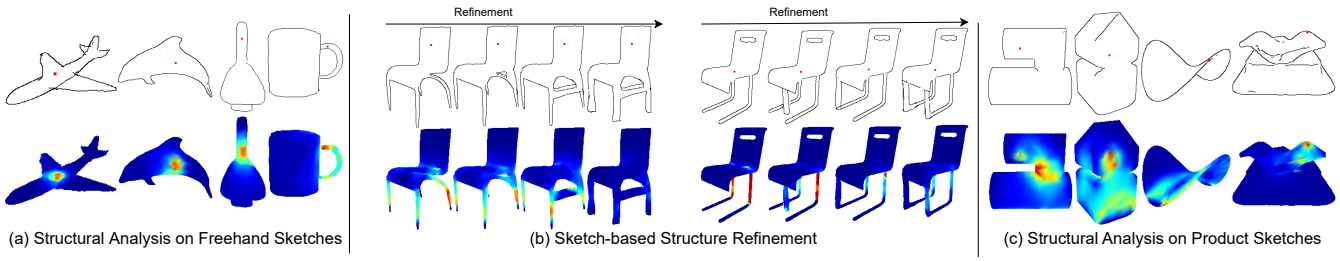

Figure 1: *Our StructureLDM enables fast, interactive structural analysis across various sketch-based application scenarios by allowing users to apply forces (red dots) directly on the sketch. (a) shows sketch-based structural analysis for identifying the weak regions (warmer colors) on the sketched structure. (b) illustrates how our StructureLDM supports users perform structural refinement on these weak regions of sketches: the upper row presents the progressively refined sketches, while the bottom row shows corresponding stress maps computed by our system. (c) demonstrates the application of our structural analysis to product sketches in the OpenSketch dataset [GSH*19].*

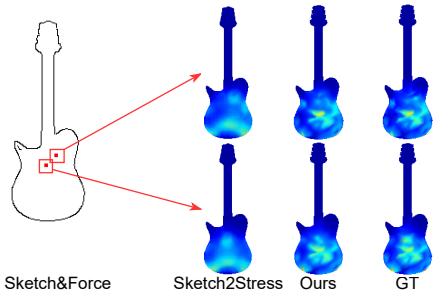

Figure 2: *Comparison on fine-grained force point perception between the previous approach (Sketch2Stress [YXLF24]) and our method. For spatially close input forces, our method can produce more distinguishable stress maps.*

Since existing digital structural analysis methods [SVB*12, LSD*16, UMK17, ZPZ13] are primarily designed for 3D prototypes, they are not directly applicable to our sketch-based structural analysis task. This limitation arises from the inherent discrepancy between the abstract 2D sketches and the precise 3D models required by these methods. Pioneering work such as Sketch2Stress [YXLF24] has explored the data-driven way, leveraging generative adversarial networks (GANs) [GPAM*14] to learn an end-to-end mapping from the input sketch and the force point map (indicating the applied force) to a stress map. While this method validated the feasibility of sketch-based structural analysis, the inherent training instability and mode-collapse issues of GANs still limit the fidelity of sketch-based structural analysis. For example, the high-stress regions in the generated stress maps are often too blurred to reliably reflect the weak areas in a sketched structure, and the current method struggles to respond accurately when the applied forces are spatially close, as shown in Figure 2.

To address these challenges, we introduce *StructureLDM*, a novel sketch-based structural analysis architecture built upon a Latent Diffusion Model (LDM) [RBL*22]. By leveraging the controllable iterative denoising process, our *StructureLDM* can synthesize stress distribution maps with significantly higher quality and richer details than GAN-based methods [IZZE17, WLZ*18]. To more precisely encode the locations of the applied force, we incorporate a position-embedding mechanism into the force representation and condition it jointly with diffusion timesteps during generation process. We further observe that although the LDM reliably reconstructs the low-frequency components of stress maps, it struggles to capture the high-frequency details. Therefore, we further introduce a Fourier-domain structural consistency loss to constrain the generation of high-frequency components of the generated stress maps, enhancing the fidelity and quality of sketch-based structural analysis.

Comprehensive qualitative and quantitative experiments demonstrate the superiority of our *StructureLDM* in terms of high-precision stress map generation and fine-grained force controllability compared with existing methods.

Our contributions can be summarized as follows:

- We introduce a latent diffusion model (LDM) to sketch-based structural analysis for generating high-fidelity and precise stress maps.
- We design a force-point embedding module (FEM) that encodes force positions (2D coordinates) into distinguishable, high-dimensional embeddings, enabling precise force-aware conditioning.
- We propose a Fourier Domain Loss (FDL) combined with a spatial-domain pixel-wise loss, enhancing the quality of the generated results in both the spatial and frequency domains.

2. Related Work

Our work is related to structural analysis (Section 2.1), sketch-based structural analysis (Section 2.2), and sketch-based image synthesis (Section 2.3). Here we only review prior works that are closely related to ours.

2.1. Structural Analysis

Numerous prior works study computational analysis of the structural soundness of 3D shapes, focusing on evaluating the behavior of fabricated structures under external applied forces. The use

of computational structural analysis spans structural weakness detection [ZPZ13, TR21, MR09], topology optimization [PRZ17], and material distribution optimization [UMK17, FZZ*20a, ZCT16, LSZ*14, DLL*15, FZZ*20b].

Telea and Jalba [TJ11] pioneered the structural analysis of 3D-printed models by identifying thin and thick parts and estimating whether a thin part could support its attached parts under several pre-defined geometric rules. Later, Stava et al. [SVB*12] adopted the finite element method to identify and strengthen fragile regions in 3D-printed models under gravity and two-finger grasping scenarios. And Zhou et al. [ZPZ13] then propose an analysis technique that predicts fragile regions under worst-case external loads by identifying structural areas that are the most likely to fail under arbitrary force configurations. Later, Langlois et al. [LSD*16] introduced a stochastic finite element method that models the mechanical response of contact regions and predicts their failure probability under the external force. Unlike aforementioned methods that target specific force conditions, Ulu et al. [UMK17] proposed a general structural optimization scheme for 3D models with arbitrary force locations. This approach computes reasonable material distributions and offers practical support for structural design.

However, these computational structural analysis methods are mainly designed for the well-defined and watertight 3D prototypes. Fundamental obstacles arise when applying them to early-stage conceptual sketches that are typically characterized by sparsity, view dependence, and geometric incompleteness [RSR*26]. Therefore, conventional structural analysis methods cannot be directly applied to our sketch-based setting. In this work, we train our *StructureLDM* on sketches from multiple views to mitigate the ambiguity caused by view dependency.

2.2. Sketch-Based Structural Analysis

Sketch-based structural analysis is challenging due to the highly abstract and ambiguous nature of input sketches [YYL*17]. In the literature, existing approaches can be broadly grouped into two branches: (i) pipeline that combines 3D reconstruction [GRYF21, ZGG21, LGL*24] and 3D structural simulation [ZPZ13, TR21, MR09, UMK17], and (ii) generative paradigm that predicts structural stress responses from sketches without explicit 3D reconstruction.

In reconstruction-and-simulation paradigm, a 2D sketch is first converted into a 3D model using single-view sketch-based 3D modeling techniques [MON*19, GRYF21, WZL*18, ZGG21, LGL*24]. The resulting geometry is then post-processed (e.g., repair, simplification) to obtain a watertight 3D model suitable for simulation. Finally, 3D structural analysis [SVB*12, ZPZ13, UMK17, PRZ17] is performed on this reconstructed model to produce the corresponding stress map. However, the faithfulness of this paradigm is fundamentally constrained by the quality of the reconstruction step since single-view reconstruction [MON*19, GRYF21, LGL*24, ZGG21, WZL*18] are still not reliable due to the lack of depth cues and the inherent ambiguity of 2D sketches. Moreover, high-precision sketches are too demanding for novice users to create [YLZ*21], which in turn decrease the reconstruction and analysis quality. For instance, ONet [MON*19] and Sketch2Mesh [GRYF21] tend to produce isolated noise, disconnected components, and geometric inconsistencies, while Pixel2Mesh [WZL*18] exhibits corrupted surface patches, and Sketch2Model [ZGG21] commonly loses geometric details due to over-smoothing. These reconstruction artifacts severely degrade the accuracy and robustness of subsequent stress simulations.

To avoid these reconstruction artifacts, a direct GAN-based structural analysis paradigm [YXLF24] has been proposed. Sketch2Stress [YXLF24] bypasses explicit 3D reconstruction by formulating the sketch-based structural analysis as an image-to-image translation problem, where a two-stage GAN-based model is introduced to map an input sketch and force conditions directly to a stress distribution. This paradigm effectively eliminates error accumulation from intermediate reconstruction processes, providing a fast and more robust alternative for sketch-level structural analysis and optimization. However, due to common GAN issues, this approach cannot guarantee the accuracy of the generated stress maps, resulting in over-smoothed stress fields with high-frequency details suppressed or lost. In this work, we leverage a latent stable diffusion model and dedicated designs for the force-point embedding module (FEM) and the Fourier-domain loss (FDL) to enhance the fidelity of sketch-based structural analysis.

2.3. Sketch-Based Image Synthesis

In sketch-based image synthesis, the underlying generative models have undergone a paradigm shift from conditional generative adversarial networks (cGANs) [IZZE17, WLZ*18, SDY*18, JCLL20, YXLF24] to diffusion models [WWY*26, SME20, HJA20, RBL*22, DZWM24, WLS*25]. The primary goal of these approaches is to strictly preserve the the input sketch structure while fully leveraging control signals to synthesize high-fidelity image content.

Early cGAN-based methods [IZZE17, WLZ*18] laid the groundwork for structured conditional generation. pix2pix [IZZE17] first enabled end-to-end mapping from sketches and semantic masks to photorealistic images. Subsequently, Su et al. [SDY*18] extended this paradigram to 3D perception by enhancing geometric understanding via normal map generation. Later, Jiao et al. [JCLL20] further incorporated category and depth cues to improve shape parsing from abstract inputs such as tactile sketches. More recently, Sketch2Stress [YXLF24] introduced a dual-branch GAN framework to simultaneously synthesize normal maps and stress maps, extending the sketch translation to physical simulated prediction. However, these cGAN-based methods can only learn a coarse distribution-level similarity via a "fool-and-detect" game between the generator and discriminator, and tend to over-smooth sharp stress boundaries and ignore rare response samples, limiting their application for sketch-based structural analysis task.

Recently, diffusion models [SME20, HJA20, DZWM24, ZHW*25, CCZ*25, DWL*25] have surpassed cGANs in both generation quality and generalization, largely thanks to their progressive denoising mechanism (recovering the fine details via iterative refinement) and large volume of training datasets. Rombach et al. [RBL*22] proposed Latent Diffusion Model (LDM) to perform diffusion process in a low-dimensional compact

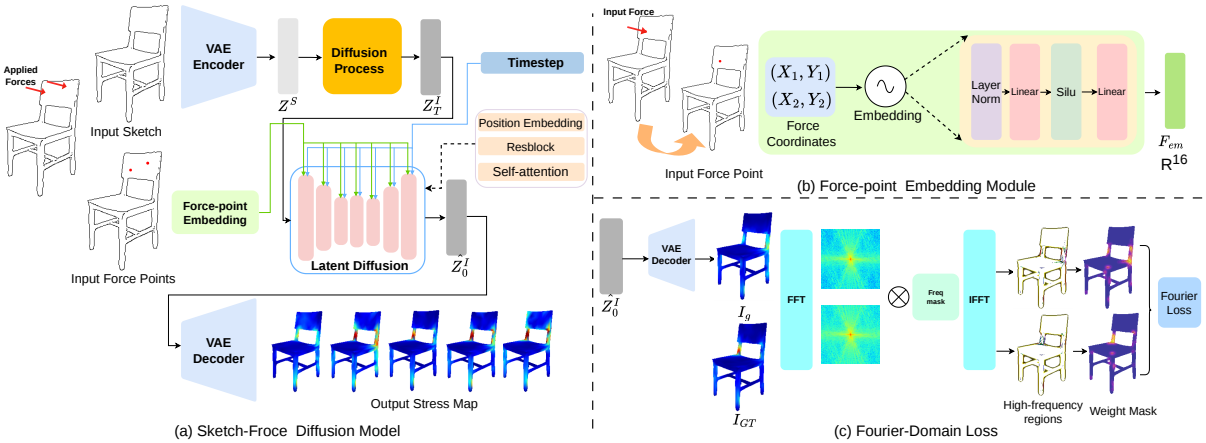

Figure 3: *The system diagram of StructureLDM. (a) shows the overall architecture: an input sketch is first encoded into a latent space via a VAE encoder and then perturbed with noise (omitted here for clarity). Meanwhile, the input force point is projected into a high-dimensional space through our force-point embedding module (FEM) and then concatenated with timestep to jointly guide the latent diffusion process. The final stress map is then reconstructed from the refined latent code by the VAE decoder. (b) illustrates the force-point embedding module (FEM), which maps 2D coordinates to a high-dimensional embedding space enabling a fine-grained, discriminative representation of the force point. (c) depicts the Fourier-domain loss (FDL), which emphasizes the high-frequency details and better aligns the generated stress map with the visual characteristics of the ground-truth stress distribution.*

latent space, significantly reducing computation while naturally supporting conditional inputs, such as sketches, masks, and other conditions, which has made LDM a commonly adopted backbone for sketch-driven generation. For example, building upon LDM, ControlNet [ZRA23] injects sketches as structural control signals into LDM via zero-convolution layers to produce high-fidelity images. Similarly, HDiff-HIR [LCF*25] incorporates physical imaging priors through a mask-conditioning module, allowing the model to better localize and guide the generation process. In this paper, we also adopt LDM as the generative backbone. We observe that relying only on LDM cannot fully capture high-frequency details in the structural stress map. Therefore, we further design a Fourier-domain loss to better model high-frequency structural details.

3. Methodology

In this section, we provide a detailed elaboration of our *StructureLDM*. We begin with the preliminary of our method (Section 3.1), followed by the introduction of the details on three main components: Force conditioned sketch latent diffusion (Section 3.2), Force embedding module (Section 3.3), and Fourier-Domain loss and objective functions (Section 3.4). Finally, we present the sketching interface specially designed for our *StructureLDM* in Section 3.5.

3.1. Preliminaries

We aim to learn an end-to-end mapping F_θ from a 2D sketch S and a user-specified force point P to the structural stress map I which describes the corresponding stress distribution over the sketched structure.

$$I = F_\theta(S, P) \tag{1}$$

Following prior works [YXLF24, UMK17], we introduce several assumptions and constraints on the external forces and material properties of the objects:

(i) External force. A single force is decomposed into three components: magnitude, direction, and location. In our setup, the force magnitude is fixed at 100 N; the force position is projected to 2D coordinates $P_{x,y}$; and the force direction is set to be opposite to the surface normal at the force point $P_{x,y}$.

(ii) Material property. We assume that the structure represented by the sketch is made of a homogeneous, isotropic, linear elastic material with small deformations [YXLF24].

These assumptions are motivated by the fact that, in early-stage design, designers typically use nominal material parameters and moderate forces to explore alternatives. In this regime, such assumptions provide sufficiently accurate stress trends while keeping the analysis simple and fast. Therefore, under these settings, the stress distribution I_g is conditionally determined by the sketched structure S and the force point P, thereby casting sketch-based structural analysis as a conditional image-to-image translation task.

3.2. Architecture of *StructureLDM*

As shown in Figure 3, the architecture of our *StructureLDM* contains three components: sketch-force diffusion module, force-point embedding module (FEM) (Figure 3 (b)), and Fourier-domain loss (FDL) (Figure 3 (c)). Given an input sketch S and a force point P, our *StructureLDM* F_θ aims to generate an accurate 2D stress map I for rapid structural analysis. To formulate this mapping function in

Equation 1, we cast the sketch-force to stress generation problem into a joint probability distribution:

$$I = F_\theta(S,P) = F_\theta(S = s \text{ and } P = p_{x,y}) \tag{2}$$

where s and $p_{x,y}$ are the concrete sample for the input sketch S and the coordinate of force point P, respectively.

Generally, we employ a latent diffusion model (LDM) as the backbone of *StructureLDM* F_θ to learn this joint probability distribution. To project the input sketch and force point into a shared latent space, we first use a Variational Autoencoder (VAE) encoder $E(\cdot)$ and the force-point embedding module (FEM) to encode the input sketch S and force point P, as shown in Figure 3 (a). The resulting latent embeddings of the sketch and force are then concatenated with the timestep and noise vector to form a unified latent representation. Denoising diffusion is subsequently performed within this compact latent space using the denoising U-Net ϵ_θ, which iteratively refines the concatenated noisy latent representation toward a clean latent code $\hat{z}_0^I$. Finally, we use the VAE decoder $D(\cdot)$ to map the generated latent code $\hat{z}_0^I$ back to the image domain to reconstruct the corresponding structural stress map I.

Directly training a model maping inputs (e.g., the sparse sketch) to the output stress map is highly challenging due to the significant domain gap between the sparse sketch and the dense stress map. To facilitate learning, we follow a two-stage training strategy as [RBL*22, PX23]: (i) Stress-map reconstruction pretraining and (ii) Sketch-to-stress generation, as shown in Figure 3 (a).

In the first stress-map reconstruction stage, we finetune the Variational Autoencoder (VAE) solely on ground-truth stress maps $I \in \mathbb{R}^{256\times256\times3}$ in a self-supervised manner, with an $\mathcal{L}_1$ loss between the reconstructed output $D(E(I))$ and the stress map I, together with a KL divergence term that prevents the model from collapsing into a naive autoencoder (Equation 3). This allows the model to learn a compact latent representation $z^I \in \mathbb{R}^{32\times32\times4}$ and construct an initial warm-up latent space for stress maps:

$$\mathcal{L}_{VAE} = ||D(E(I)) - I||_1^1 + \alpha \cdot \mathcal{L}_{KL} \tag{3}$$

where $E(\cdot)$ and $D(\cdot)$ are the VAE encoder and decoder, respectively, while $\alpha = 1e^{-6}$ and represents the balancing weight of the KL divergence term $\mathcal{L}_{\text{KL}}$.

In the second sketch-to-stress generation stage, our goal is to generate the stress map conditioning on the input sketch and force point. Therefore, unlike the original LDM [RBL*22], which discards the VAE encoder $E(\cdot)$ and retains only the decoder $D(\cdot)$ for image generation, our *StructureLDM* preserves both components and uses the encoder to project the input sketch into the stress-map latent space as:

$$z^S = E(S) \tag{4}$$

where S and z^S denote the input sketch and its latent embedding in the stress-map latent space, respectively. This design guarantees a tight correspondence between the output stress map and the input sketch in our empirical experiments. Conditioned on the latent embedding of input sketch z^S, and the input force point feature z_{fp} (extracted via FEM Figure 3 (b)), we train a customized denoising U-Net $\epsilon_\theta(\cdot)$ to perform the diffusion and reverse denoising process,

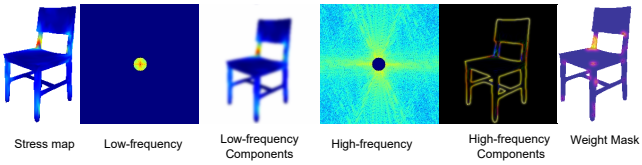

Figure 4: *Fourier frequency decomposition and high frequency (stress) region localization.*

producing the latent code $\hat{z}_0^I$ of the corresponding stress map iteratively. The training objective is formulated as:

$$\mathcal{L}_{diff} = \mathbb{E}_{\mathbf{z}_0^I,t,z^S,z_{fp},\epsilon}\left(\left\|\epsilon - \epsilon_\theta\left(\mathbf{z}_t^I, t, z^S, z_{fp}\right)\right\|_2^2\right) \tag{5}$$

Here, z_0^I is the latent code of the ground-truth stress map, and z_t^I denotes the intermediate latent code at timesetep t.

After obtaining the denoised latent code $\hat{z}_0^I$, we employ the VAE decoder $D(\cdot)$ to decode $\hat{z}_0^I$ to the stress map, as shown in Figure 3 (a). We regularize the generation process with an $\mathcal{L}1$ loss function:

$$\mathcal{L}_{regu} = \|I_g - I_{gt}\|^1, \quad I_g = D(\hat{z}_0^I) \tag{6}$$

where I_g and I_{gt} denote the generated and ground truth stress map, respectively. Users can perform structural analysis on the sketched structures under any arbitrary applied forces, as illustrated in Figure 3 (a).

3.3. Force-Point Embedding Module (FEM)

The accurate prediction of stress distributions critically depends on the distinct representation and effective integration of force point information, especially among spatially proximal points, within the diffusion generation process. The prior approach [YXLF24] represents force points as a point map with the same spatial resolution as the input sketch. However, such representations are prone to being attenuated through successive convolutional and downsampling layers, leading to the loss of precise force-location information and reduced fidelity in high-stress region generation.

To mitigate this issue, we introduce a force-point embedding module (FEM) that encodes force points in a more robust and precise manner, as shown in 3 (b). The core idea of this module is to explicitly encode discrete force-point coordinates into a structured high-dimensional vector and to provide precise guidance, which fuses this embedding with the timestep information. The precise integration of force information is organized in three steps, as illustrated in Figure 3 (b):

(i) Normalization: We first locate the pixel position (x,y) of the force point $P_{x,y}$, and normalize it to [-1, 1] to obtain $\mathbf{P}_{\mathbf{x_n},\mathbf{y_n}} = (x_n, y_n) \in \mathbb{R}^2$;

(ii) Embedding: We then embed P_{x_n,y_n} into a high-dimensional vector z_{fp} using a two-layer Multi-Layer Perceptron (MLP) with SiLU activation function, which is expressed as follows:

$$z_{fp} = W_2 \cdot \text{SiLU}\left(W_1 \cdot \mathbf{P}_{\mathbf{x_n},\mathbf{y_n}} + b_1\right) + b_2 \tag{7}$$

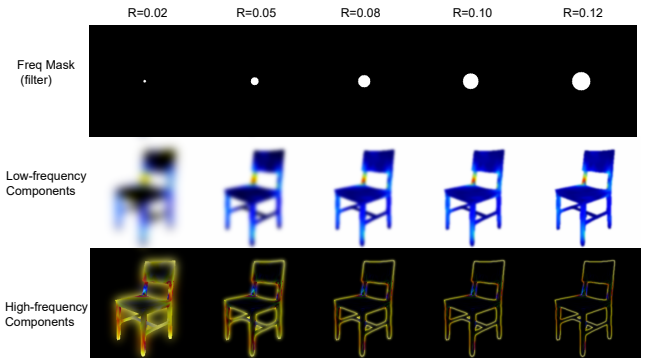

Figure 5: *Fourier frequency decomposition under varying radius of Freq-mask.*

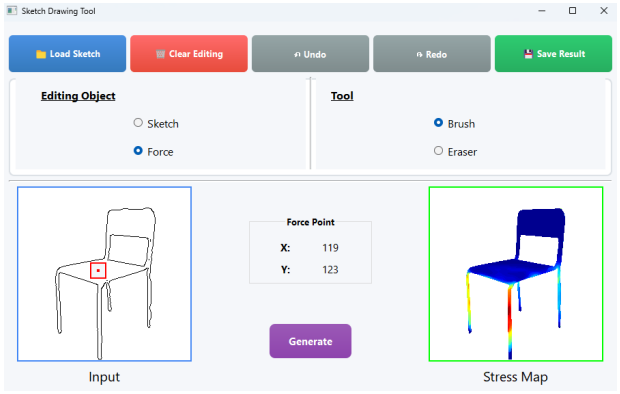

Figure 6: *Interface for sketch-based structural analysis.*

where $W_1 \in \mathbb{R}^{16\times 2}$, $b_1 \in \mathbb{R}^{16}$, $W_2 \in \mathbb{R}^{16\times 16}$, and $b_2 \in \mathbb{R}^{16}$ are trainable parameters;

(iii) Fusion: After obtaining z_{fp}, we fuse it with the timestep embedding z_t through an element-wise modulation operation [PSdV*18].

In this way, our *StructureLDM* injects the precise force guidance into the generation process by jointly conditioning on temporal information and force-point coordinate information.

3.4. Fourier-Domain Loss (FDL) and Training Objective

To ensure that the generated stress map I_g reflects both high spatial fidelity and physical plausibility, we design a multi-domain objective function for optimization.

3.4.1. Fourier-Domain Loss (FDL)

Depending solely on the spatial-domain loss functions [YXLF24] can not accurately reconstruct the brittle regions in stress maps, which typically appear as sharp stress jumps and high-gradient patterns (bright red and yellow areas), as illustrated as high-frequency components in Figure 4. These regions are difficult to depict reliably using simple thresholds or global spatial statistics.

To address this issue, we introduce a Fourier-domain decomposition method for precise localization and explicit enhancement of high-frequency regions, as illustrated in Figure 4. The generated stress map is first transformed to the frequency domain via a Fast Fourier Transform (FFT), where adaptive high-pass filters (*Freq mask* in Figure 3 (c)) are derived from the spectral energy distribution to isolate high-frequency components. We further visualize a spatial weight mask for the corresponding high-frequency components in Figures 3 (c) and 4.

We show the effects of different *Freq mask* sizes on Fourier-domain decomposition and the corresponding separated stress regions using an inverse fast Fourier transform (IFFT) in Figure 5, where R denotes the radius of *Freq mask*. We can see that when R is small, excessive low-frequency content is wrongly assigned to the high-frequency band; as R increases, the low-frequency components become clearer, and the high-frequency components gradually concentrate around sharp stress jumps with high gradient patterns. In our work, we adopted the empirical value $R = 0.08$, to ensure the clear separation of the low-frequency and the high-frequency regions.

Given the generated stress map I_g and the ground-truth I_{gt}, we compute the channel-wise Fourier-domain loss as:

$$\mathcal{L}_{\text{fourier}} = \frac{1}{3HW}\sum_{c=1}^{3}\sum_{i,j} W_f(i,j)\cdot\left|\left|\text{FFT}\left(I_g^c\right)\right| - \left|\text{FFT}\left(I_{gt}^c\right)\right|\right| \quad (8)$$

where H and W are the size of stress maps, c represents the image channels, and $W_f(\cdot)$ is the threshold weights to isolate and enhance the high-frequency components. The frequency-domain threshold matrix $W_f(\cdot)$ is defined as:

$$W_f(i,j) = \frac{\max(d(i,j)-R,0)}{\max(d)-R} \quad (9)$$

Here, $d(i,j)$ represents the distance from the current frequency point (i,j) to the center of the spectrum.

3.4.2. Overall Objective Function

The final training objective $\mathcal{L}_{total}$ is the weighted sum of the diffusion loss $\mathcal{L}_{diff}$, the reconstructed loss $\mathcal{L}_{regu}$, and the Fourier-domain loss $\mathcal{L}_{fourier}$:

$$\mathcal{L}_{total} = \beta\cdot\mathcal{L}_{diff} + \gamma\cdot\mathcal{L}_{regu} + \lambda\cdot\mathcal{L}_{fourier} \quad (10)$$

where $\beta = 1.0$, $\gamma = 1.0$, and $\lambda = 0.15$, are the hyperparameters to balance each loss term. After training, the input sketch and force can be efficiently compressed into a bounded, compact and distinctive latent space without compromising important information. Also, our *StructureLDM* model can precisely follow the sketch input and force guidance under various designed structures, enabling more fine-grained and detailed controlled stress map generation. We show several generated results in Figure 7.

3.5. Sketching Interface

To apply our *StructureLDM* to the practical design workflows and validate its effectiveness, we further develop an interactive sketch-

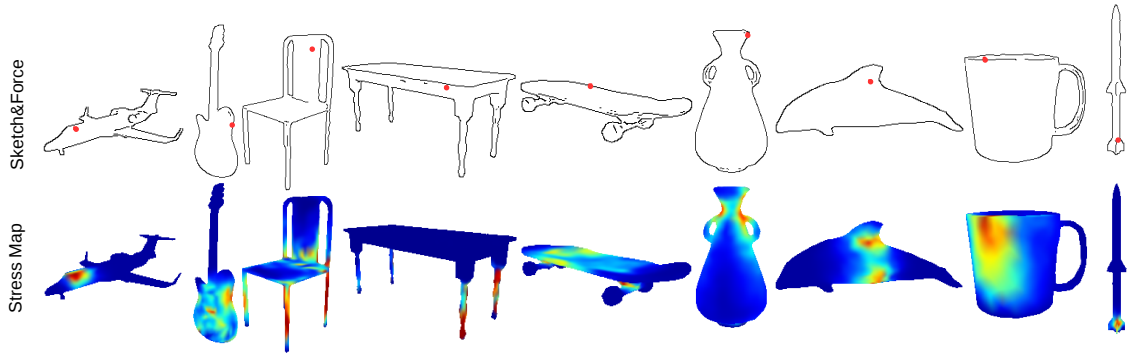

Figure 7: *Nine representative categories in our collected dataset. The top row displays the input sketches with the applied external force marked by red dots, while the bottom row shows the generated stress maps with our StructureLDM.*

ing interface for sketch-based structural analysis and refinement, as illustrated in Figure 6. Our interface offers two modes, ***Sketch*** mode and ***Force*** mode, with switching enabled via radio buttons (Figure 6). In *Sketch* mode, users can create a 2D sketch either by freehand drawing or importing an existing sketch and editing it with our editing functions (e.g., erase, undo, redo). After completing the sketch, users can switch to *Force* mode to apply the external forces by directly clicking on the desired regions on of the sketch. Finally, by clicking on the ***Generate*** button, our *StructureLDM* will run automatically in the background to produce the corresponding structural stress map.

Based on the generated structural stress map, users can switch back to *Sketch* mode to further refine the designed structure. Through this rapid "sketching-analyzing-refining" iteration using our sketching interface, users can effectively identify structural weakness and improve the structural soundness of the sketched structure.

4. Experiment

To comprehensively evaluate our proposed *StructureLDM* on the sketch-based structural analysis task, we conduct both qualitative (Section 4.3) and quantitative (Section 4.4) experiments to examine the quality of generated results.

4.1. Benchmark

Datasets. We follow the same strategy as [YXLF24] to prepare our sketch-force-stress dataset. Slightly different from the dataset in [YXLF24], we further filtered out the repeated shapes to increase the diversity and challenge of the geometry and structure of the input sketches. Note that we split our training and testing data at the category level, ensuring that some shapes within each category remain unseen for the training model.

More specifically, we construct the dataset by simulating forces on normalized, upright 3D models with the bottom on the ground plane and upper contact regions corresponding to the resting regions. We uniformly sample forces over these contact regions and simulate them to obtain full 3D stress distributions. From these simulations, we then render multi-view outputs at predefined viewpoints, including 2D projections, force maps, and stress maps. Note that the corresponding sketches are generated using Canny edge detection, yielding paired sketch–force–stress supervision, which is a common practice in sketch-based learning due to the scarcity of large-scale hand-drawn data. The statistics of our collected dataset are detailed in Table 1.

Table 1: *Data distribution of our collected sketch-force-stress dataset.*

Category	#Shape	#Views	#Sketches	#Force-points	#Stress-map
Chair	2,161	3	6,483	773,938	773,938
Table	2,378	2	4,756	683,219	683,219
Airplane	2,231	3	6,693	403,926	403,926
Vase	184	1	184	22,824	22,824
Mug	164	1	164	15,840	15,840
Skateboard	134	3	402	24,471	24,471
Rocket	49	1	49	4,291	4,291
Guitar	39	2	78	9,186	9,186
Fish	20	1	20	2,640	2,640

Evaluation Metrics. We adopt four widely adopted metrics [YXLF24] for performance comparison: Mean Absolute Error (MAE) and F-Measure (FM) quantify the pixel-wise stress accuracy; Earth Mover's Distance (EMD) and Fréchet Inception Distance (FID) evaluate the collective discrepancy of the generated images.

Baseline Methods. One direct baseline in sketch-based structural analysis is Sketch2Stress [YXLF24]. Besides, we compare *StructureLDM* with representative sketch-based image synthesis approaches. pix2pix [IZZE17] and pix2pixHD [WLZ*18] serve as the primary baselines, and ControlNet [DZWM24] represents the most-recent conditional generation paradigm with diffusion model as the backbone. Note that all baselines use their officially released architectures, with only the input/output channels adjusted to fit our task. Specifically, ControlNet is built on Stable Diffusion v1.5, taking a 2-channel input (sketch + force map), where the original pre-trained weights are frozen and only the ControlNet branch is trained. And all the compared methods are retrained on our collected dataset for a fair comparison.

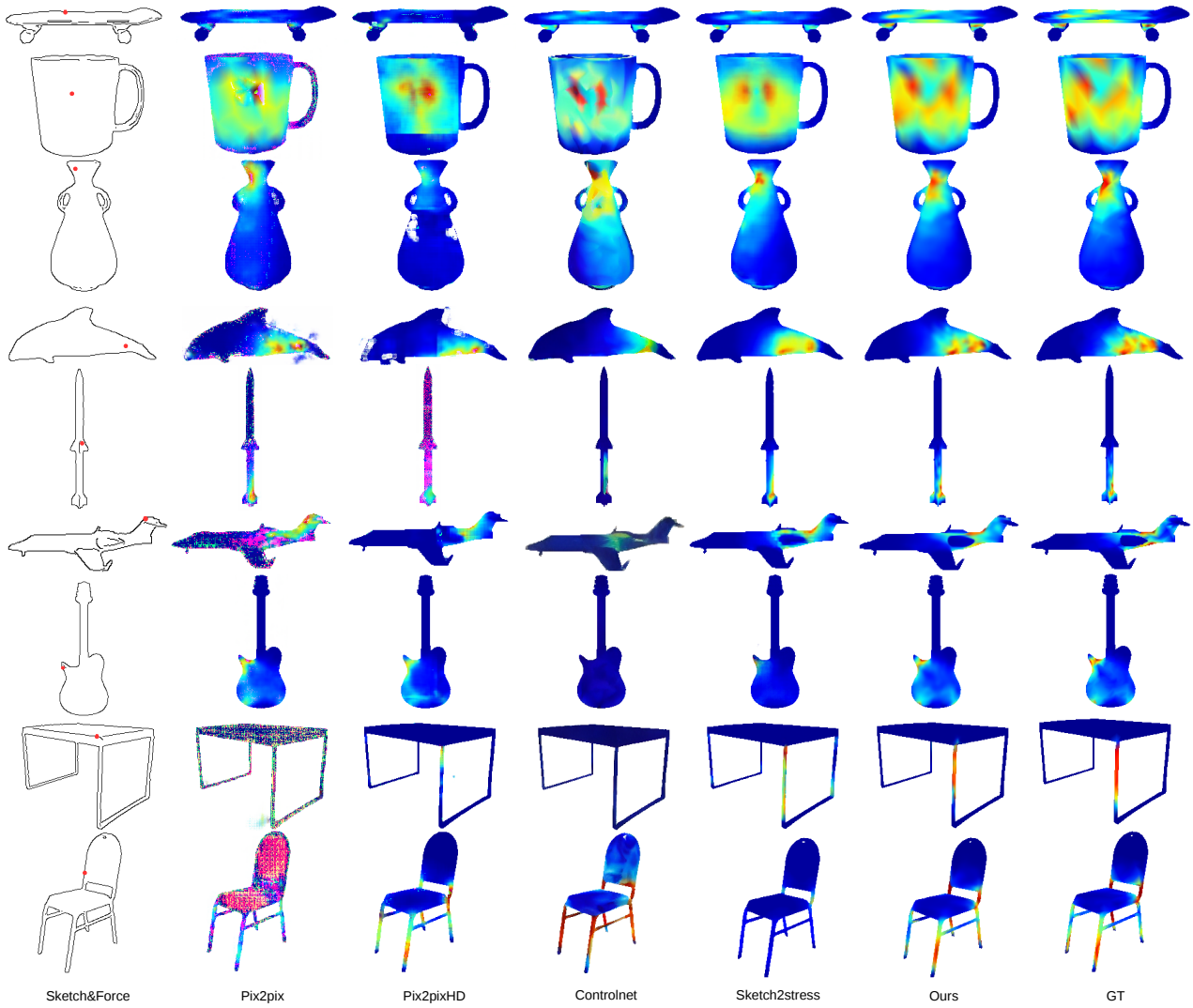

Figure 8: *Qualitative comparison of different approaches. The leftmost column shows the input sketches and applied force, and the remaining columns present the generated stress maps and the corresponding ground-truth results, respectively.*

4.2. Implementation Details

The entire model of our *StructureLDM* was trained with one NVIDIA A100 GPU. For the VAE-training, we adopted the VAE architecture of Stable Diffusion v1.4 and fine-tuned it on our dataset. During the fine-tuning phase, all input images were resized and uniformly normalized to 256×256, and all VAE parameters were updated during fine-tuning. We use the AdamW optimizer with an initial learning rate set to 1×10^{-4} and a weight decay of 1×10^{-2}. While for the latent diffusion training, we trained the denoising UNet with a batch size of 16. We employed AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$ with an initial learning rate 1×10^{-4}). After training, our *StructureLDM* spent approximately 0.3 seconds on average to generate the corresponding structural stress map from an input sketch under a specified force condition, compared with 1.4 seconds for ControlNet under the same settings.

4.3. Qualitative Evaluation

In Figure 8, we show the stress maps generated given input sketches and force points by pix2pix, pix2pixHD, ControlNet, Sketch2Stress, and our *StructureLDM* side by side for direct visual comparison. In comparison with the ground-truth, we can see that our method achieves the best performance among all the compared methods across different categories with diverse geometry. And Sketch2Stress produces the second best performance, but it tends to lose the stress details around the high-stress regions. We attribute our superiority to: (i) The fine-grained perceptive force-point embedding module (FEM), which encodes the input force points to discriminative high-dimension embeddings rather than using the simple point map representations in Sketch2Stress; (ii) The high-frequency weighted Fourier-domain loss (FDL), which emphasizes the high-stress regions and maintains the overall structural con-

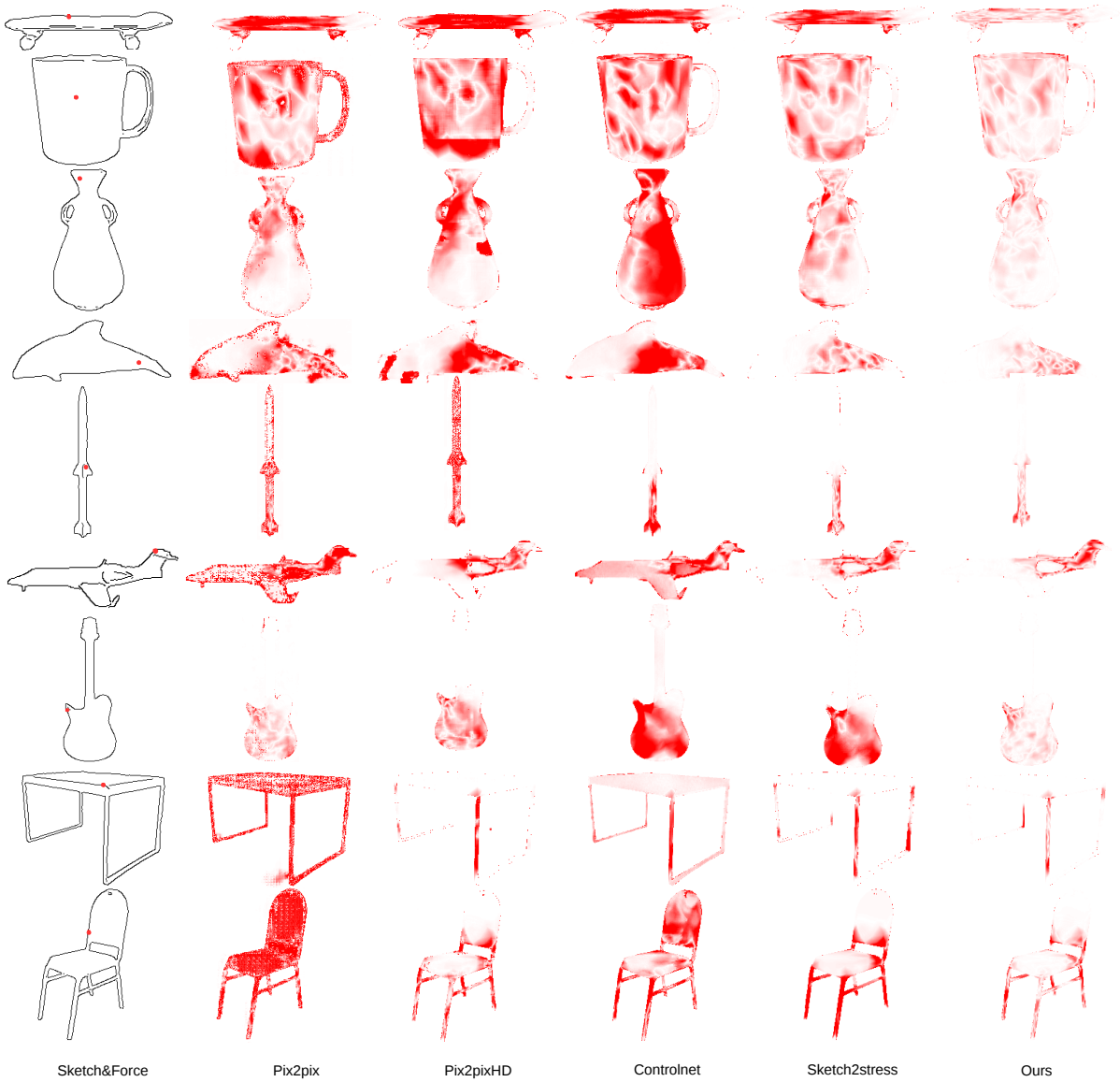

Figure 9: *Error maps of different approaches (visualizing the deviation between the generated and ground-truth stress maps). Less saturated red and lighter colors indicate smaller deviation from ground-truth. Please also examine the difference in Figure 8.*

sistency by penalizing frequency-domain discrepancies; and (iii) the generative capability of the latent diffusion model (LDM), but simply applying the LDM to our task is still insufficient, see the generated results of ControlNet in Figure 8. Although ControlNet adopts the latent diffusion model as ours, it struggles to faithfully reconstruct accurate stress distributions. without our specifically-designed modules (e.g., FEM and FDL). When it comes to pix2pix and pix2pixHD, the generated stress maps from these two methods show noticeable artifacts such as heavy noise.

To more clearly reveal the differences of the compared approaches, we compute and display the error maps between each generated stress map and the ground-truth in Figure 9. More specifically, we perform pixel-wise subtraction, normalize the deviation, and render it in red intensity, where the red channel corresponds to the error of the generated stress map, and white is zero error. As shown in Figure 9, the largest errors (deep red) typically appear along structural boundaries and at the sudden change of stress transitions, which correspond to the high-stress regions. These error maps in Figure 9 more clearly demonstrate the superiority of our proposed *StructureLDM* over all compared methods, particularly in accurately synthesizing high-stress regions, where our method exhibits the lightest red colors (i.e., the smallest errors). As the same as the qualitative results in Figure 8, Sketch2Stress ranks second on the error maps of the generated stress map, while pix2pix,

Table 2: *Quantitative comparison of different methods.*

Category	Method	MAE↓	EMD↓	FID↓	FM↑
Fish	pix2pix	9.043	6.977	203.022	0.046
	pix2pixHD	3.762	0.776	118.475	0.246
	ControlNet	5.364	0.496	38.903	0.275
	Sketch2Stress	1.743	0.155	23.135	0.553
	Ours	**1.251**	**0.129**	**17.958**	**0.567**
Guitar	pix2pix	1.887	4.174	39.269	0.064
	pix2pixHD	2.035	0.304	52.399	0.297
	ControlNet	3.307	0.333	27.979	0.282
	Sketch2Stress	1.714	0.188	16.301	0.349
	Ours	**0.675**	**0.084**	**14.359**	**0.680**
Mug	pix2pix	11.452	3.560	106.876	0.142
	pix2pixHD	8.716	0.691	61.842	0.431
	ControlNet	16.026	1.090	35.227	0.288
	Sketch2Stress	5.578	0.366	26.357	0.565
	Ours	**2.137**	**0.184**	**11.589**	**0.684**
Rocket	pix2pix	4.176	7.504	266.441	0.016
	pix2pixHD	4.514	7.669	340.108	0.007
	ControlNet	2.338	0.426	46.175	0.252
	Sketch2Stress	0.762	0.135	21.177	0.549
	Ours	**0.393**	**0.075**	**6.081**	**0.638**
Vase	pix2pix	3.747	1.079	122.408	0.169
	pix2pixHD	4.065	0.406	67.525	0.286
	ControlNet	8.381	0.747	45.354	0.231
	Sketch2Stress	1.927	0.206	27.136	0.476
	Ours	**1.435**	**0.112**	**14.306**	**0.610**
Skateboard	pix2pix	0.953	1.727	73.901	0.077
	pix2pixHD	1.749	0.697	99.910	0.114
	ControlNet	2.624	0.211	43.598	0.218
	Sketch2Stress	0.912	0.091	32.962	0.409
	Ours	**0.540**	**0.057**	**27.151**	**0.596**
Airplane	pix2pix	2.548	0.776	249.077	0.056
	pix2pixHD	0.820	0.107	38.967	0.437
	ControlNet	2.012	0.214	43.338	0.208
	Sketch2Stress	0.791	0.084	43.686	0.477
	Ours	**0.401**	**0.061**	**13.402**	**0.563**
Table	pix2pix	11.592	2.051	180.479	0.004
	pix2pixHD	0.412	0.036	12.882	0.007
	ControlNet	2.750	0.159	15.8946	0.007
	Sketch2Stress	0.396	0.017	9.974	0.008
	Ours	**0.124**	**0.004**	**1.638**	**0.027**
Chair	pix2pix	13.047	3.495	153.776	0.073
	pix2pixHD	3.398	0.565	74.690	0.418
	ControlNet	5.342	0.870	32.275	0.310
	Sketch2Stress	3.712	0.660	38.636	0.374
	Ours	**1.588**	**0.306**	**11.646**	**0.639**

pix2pixHD, and ControlNet achieve worse performance generally, prone to generating significant errors (deep red) in the high-stress regions.

4.4. Quantitative Evaluation

We report the quantitative evaluation results of the compared methods, namely, pix2pix, pix2pixHD, ControlNet, Sketch2Stress, and our proposed *StructureLDM*. In Table 2, we can observe that under four evaluation metrics, our proposed *StructureLDM* achieves the best performance (i.e., on FID and EMD metrics), achieving significant improvements in pixel-level stress accuracy (i.e., on MAE and FM metrics) compared to the competitors across all nine categories. Compared to our *StructureLDM*, pix2pix and pix2pixHD methods tend to produce stress maps that contain too much high-frequency noise (see the qualitative results in Figure 8), leading to its poor

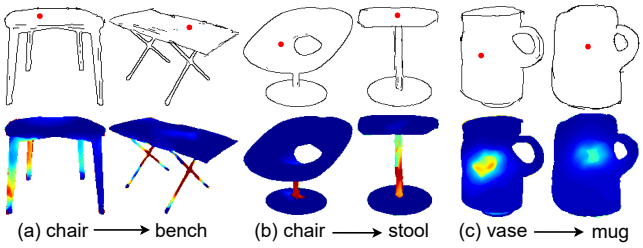

Figure 10: *Cross-category generalization. We use models trained on chairs and vases to generalize to previously unseen freehand sketches of benches, stools, and mugs.*

Table 3: *Quantitative evaluation of our ablated methods.*

Category	Ablated methods	MAE ↓	EMD ↓	FID ↓	FM ↑
Guitar	only LDM	0.774	0.100	14.944	0.644
	LDM+FDL	0.626	0.090	**13.612**	0.659
	LDM+FEM	0.729	0.095	14.447	0.661
	Full	**0.675**	**0.084**	14.359	**0.680**
Mug	only LDM	3.459	0.268	12.581	0.557
	LDM+FDL	3.092	0.242	12.407	0.590
	LDM+FEM	2.281	0.187	12.725	0.666
	Full	**2.137**	**0.184**	**11.589**	**0.684**
Airplane	only LDM	0.630	0.086	19.182	0.436
	LDM+FDL	0.605	0.083	17.610	0.443
	LDM+FEM	0.455	0.065	15.513	0.528
	Full	**0.401**	**0.061**	**13.402**	**0.563**

performance in metrics associated with image generation quality (higher EMD and FID) and pixel-wise stress accuracy (lower MAE and FM). While Sketch2Stress struggles to accurately predict stress regions surrounding the high-stress regions, resulting in compromised image quality (FID and EMD values) and pixel-wise stress accuracy (MAE and FM values). Finally, a straightforward application of the latent diffusion model in ControlNet does not perform well in our scenario in terms of image quality and pixel-wise stress accuracy.

We are also curious about how our *StructureLDM* performs on the unseen categories via free sketching. Therefore, we directly apply our well-trained *StructureLDM* to novel categories with similar structures, as shown in Figure 10. We can observe that our method is robust to moderate geometry distortion in freehand sketches and predicts feasible structural stress for unseen object categories.

4.5. Ablation Study

To evaluate the effects of the key components in our approach, namely, latent diffusion model (LDM), force-point embedding module (FEM), and Fourier-domain loss (FDL), we present both the quantitative comparison of the ablation results in Table 3 and the qualitative comparison in Figure 11. Our ablation experiments were conducted on three representative categories: guitars, mugs, and airplanes, due to their high variance of shape diversity and complexity.

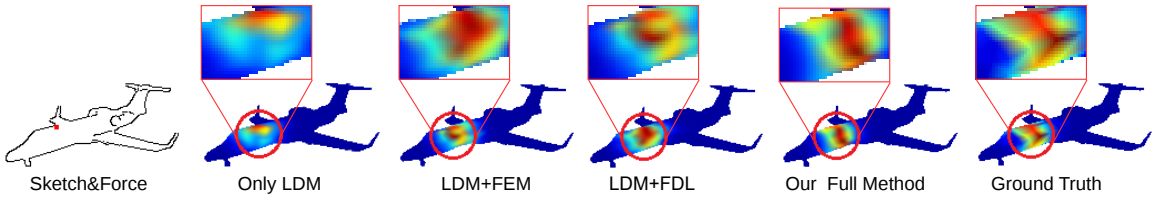

Figure 11: *Qualitative comparison of our ablated methods where we gradually add our designed LDM, FEM, and FDL, respectively.*

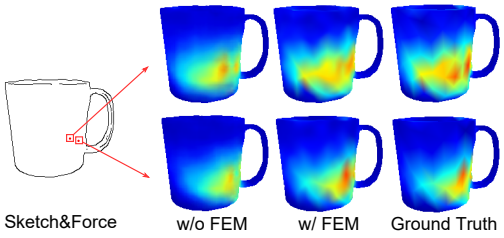

Figure 12: *Qualitative comparison of our StructureLDM with vs. without force-point embedding module (FEM) when inputting with spatially close forces.*

In Figure 11, we can observe that our full method achieves the best performance in the vast majority of cases, particularly for the complex airplane category, where it attained the top scores across all four metrics, clearly demonstrating the overall superiority of our *StructureLDM*. More in-depth analysis highlight the roles of each component as follows: (i) FDL is critical in improving the pixel-wise stress accuracy. Removing it will decrease the reconstruction accuracy (MAE and FM values) across all categories; (ii) FEM mainly enhances pixel-wise stress accuracy in high-stress regions and the fidelity of the generated results (FID and EMD values), as also illustrated in Figure 12; (iii) FEM and FDL exhibit a clear synergistic enhancement effect: when they are removed, the remaining LDM yields the worst performance across all categories and most metrics. Note that in the LDM and LDM+FDL variants without FEM, the force point is represented as a point map (following [YXLF24]), i.e., a 256×256 binary image where the force location is set to 1 and all other pixels are 0.

In Figure 12, we further highlight the importance of our FEM module under the challenging case where nearby force locations produce noticeably different stress responses. With the proposed FEM module, our method can more accurately capture the differences between the applied forces and generate stress maps that closely match the ground truth patterns.

From Table 3, we can observe that the latent diffusion model (LDM) forms the base of our *StructureLDM*. Based on LDM, compared with removing FDL, ablating the FEM component causes the larger performance drop in most cases (across four metrics and four categories), indicating that FEM contributes the most to the overall improvement. Furthermore, as shown in Figure 12, when applying spatially close forces to the same sketch, the ablated method without the FEM module cannot distinguish these close force inputs and therefore cannot faithfully produce accurate stress maps compared to the ground truth. In contrast, our method with FEM accurately captures the differences between the applied forces and generates stress maps that closely match the ground truth patterns. In summary, the ablation experiments validate and highlight the effectiveness of the key components in our *StructureLDM*.

5. Conclusion and discussion

This paper presents *StructureLDM*, a novel sketch-based structural analysis framework built upon the latent diffusion model. Without requiring 3D reconstruction, our method directly takes sketches and force points as input to generate high-fidelity stress maps, providing a fast and interactive structural evaluation tool for the early design stage. Our *StructureLDM* framework can be integrated as an early-stage structural analysis component into various CAD systems. The force point embedding module (FEM) plays a critical role in achieving fine-grained spatial force control, and its design philosophy can also be extended to other generation tasks that demand point-wise conditional control. The Fourier-domain loss (FDL) effectively improves the reconstruction quality of high-frequency details in stress maps. This module is simple yet effective and can be further generalized to tasks that require high fidelity to fine details, such as sharp-edge image generation and medical image synthesis.

While our proposed method achieves significant improvements on evaluation metrics, some limitations still exist. First, this work inherits the basic assumptions from Sketch2Stress [YXLF24] including fixed material properties, single-view, and single force inputs, whereas real engineering scenarios often require multiple forces and varying material properties. Our future work may extend the force point embedding module (FEM) to support multi-force inputs and incorporate a multi-view mechanism [YLZ*21] to strengthen 3D structural reasoning. Second, since the training data in our work are mainly derived from high-quality sketches after standardized processing, the generation quality of the proposed method for hand-drawn sketches with repeated strokes, exaggerated structures, or perspective distortion remains limited. For this issue, integrating sketch/shape beautification techniques [YLGF24, YGY*26] as a preprocessing step could improve robustness to such non-ideal inputs. Third, some of the ambiguities in our generated results could be resolved by incorporating additional normal or depth information, which we leave as a promising direction for future work. Finally, the current method is based on the latent diffusion model (LDM), whose iterative denoising process causes an inference-time efficiency bottleneck. We believe more advanced generative frameworks such as flow matching [AVE22, LGL22, LYY*25] offer potential to accelerate infer-

ence while maintaining the high-fidelity generation of stress maps, which leaves an attractive avenue for future research.

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